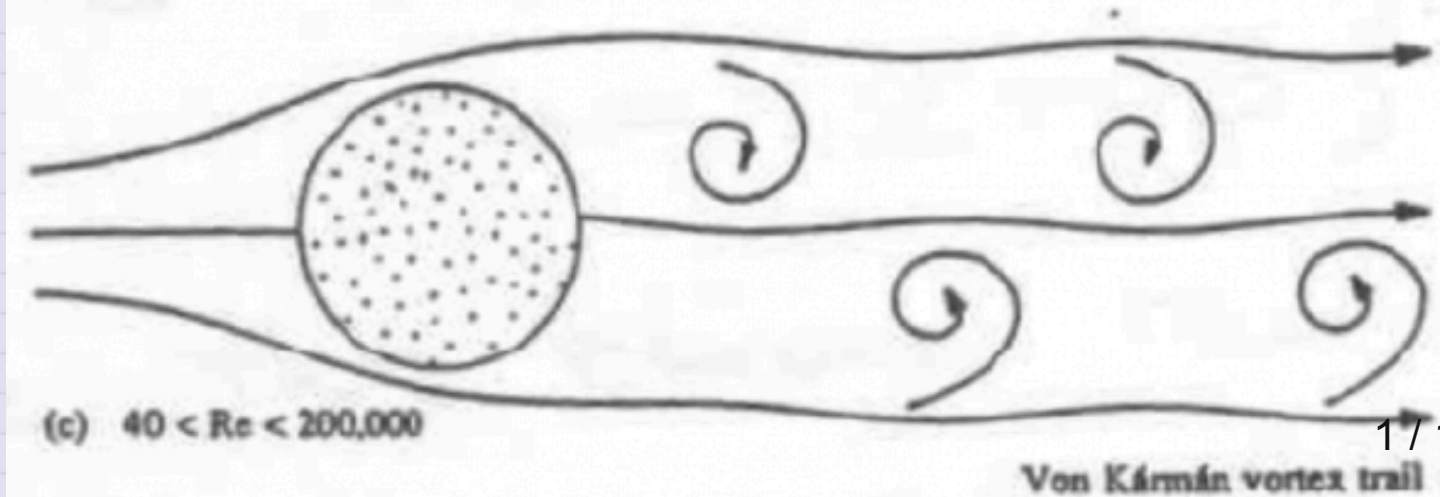
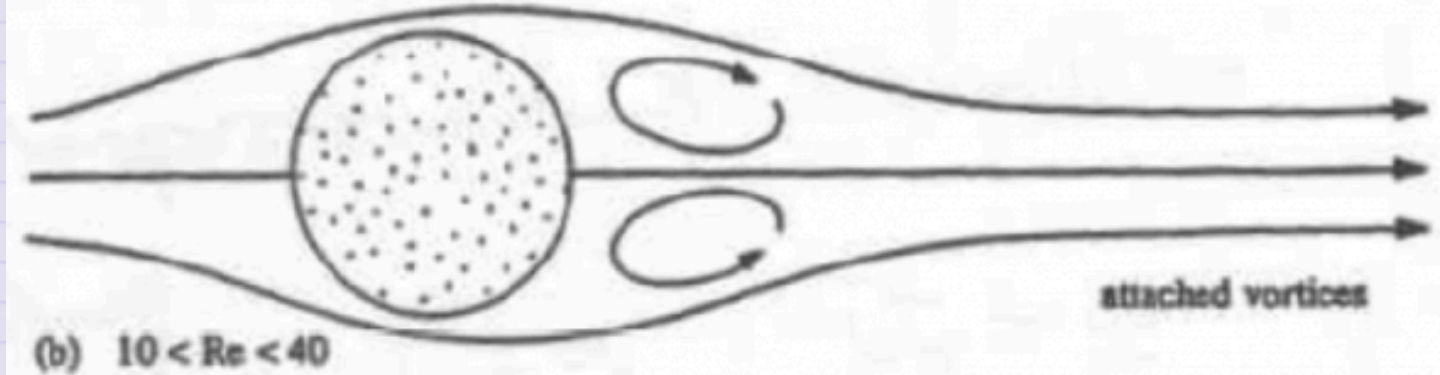
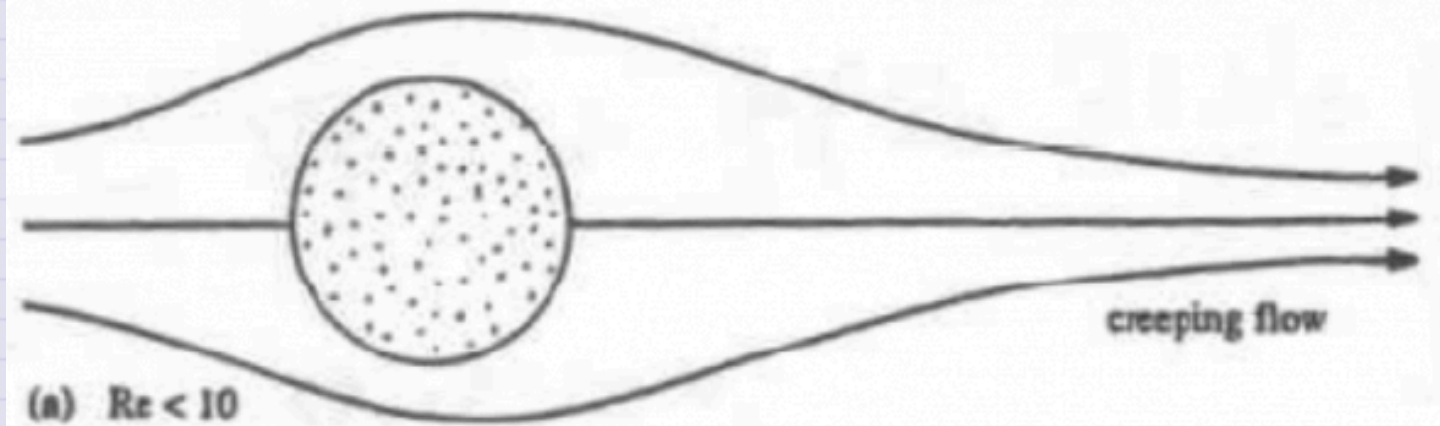


Day 06 - Making Classical Models



Plane Polar Coordinates Warm-Up

We introduced plane polar coordinates (r, ϕ) . For any position vector, $\vec{R}$, we can write:

$$\vec{R} = |\vec{R}| \hat{r} = r \hat{r}$$

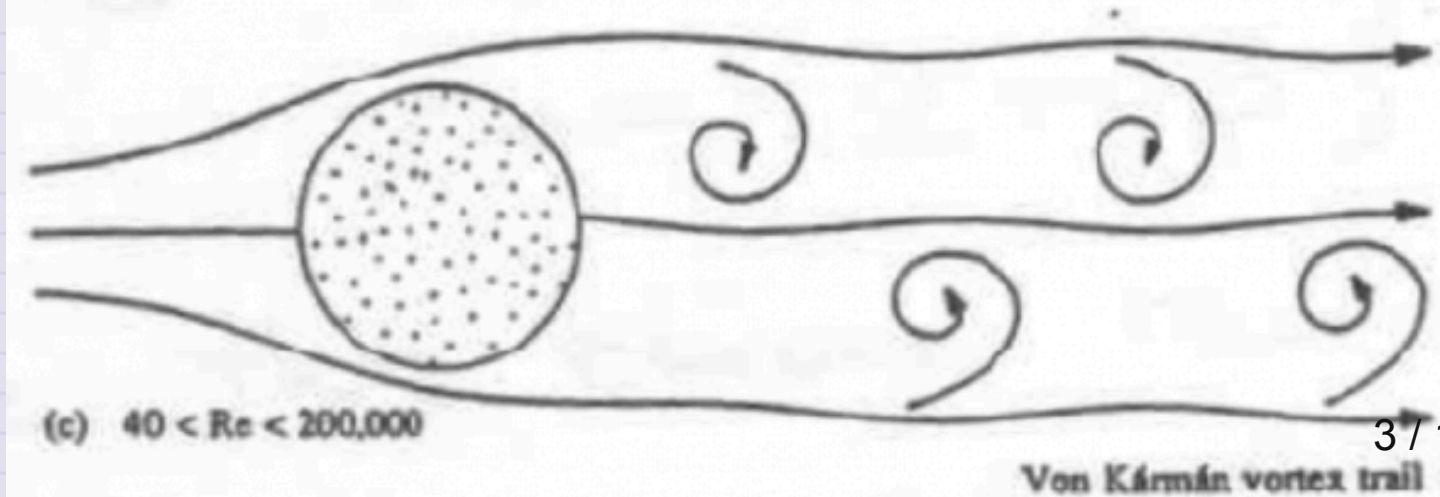
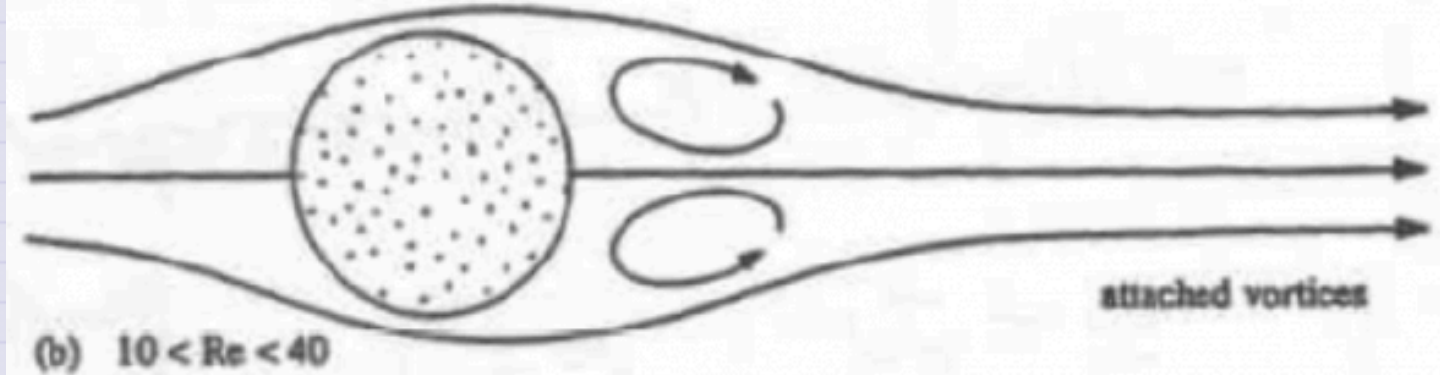
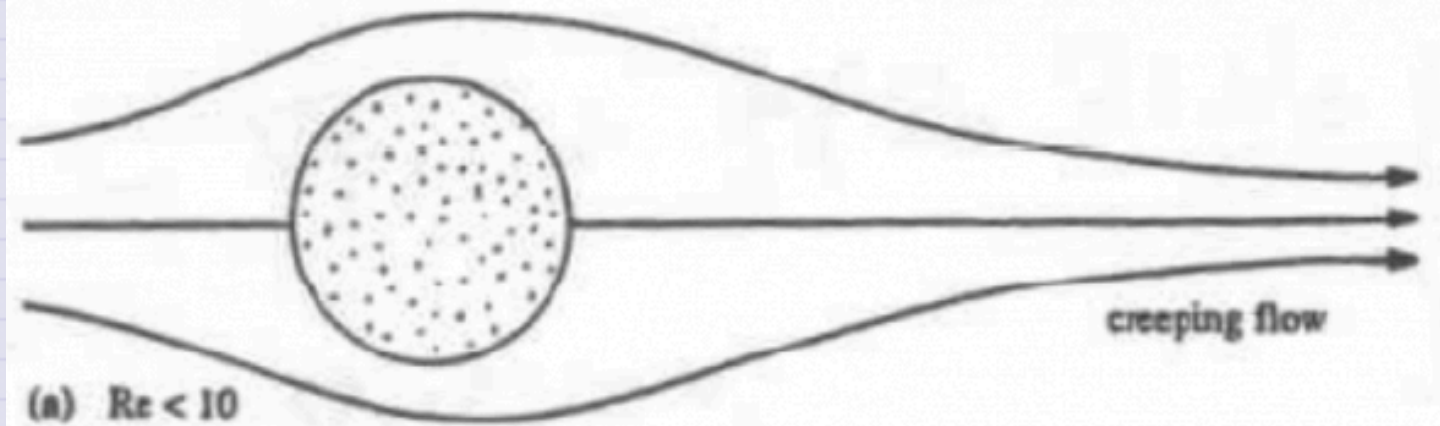
where r is the magnitude of $\vec{R}$, and $\hat{r}$ is the radial unit vector.

Find $\dot{\vec{R}} = \frac{d\vec{R}}{dt}$. Get as far as you can. Our answer will be in terms of $\hat{r}$ and $\hat{\phi}$.

Remember the chain rule and Cartesian unit vectors are fixed in space/time

$$\begin{aligned} \hat{r} &= \cos(\phi) \hat{x} + \sin(\phi) \hat{y} & \hat{\phi} &= -\sin(\phi) \hat{x} + \cos(\phi) \hat{y} \\ \frac{d}{d\phi} \cos \phi &= -\sin \phi & \frac{d}{d\phi} \sin \phi &= \cos \phi \end{aligned}$$

Day 06 - Making Classical Models



Announcements

- Homework 2 is due Friday
- Video recordings have continued to fail.
- Updated office hours:

Goals for Week 3

- Be able to answer the following questions.
 - What is Mathematical Modeling?
 - What is the process for analyzing these models?
- Be able to solve "Simple" Motion Problems with Newton's Laws.

Modeling Video



Source: <https://www.youtube.com/watch?v=dkTncoPqo5Y>

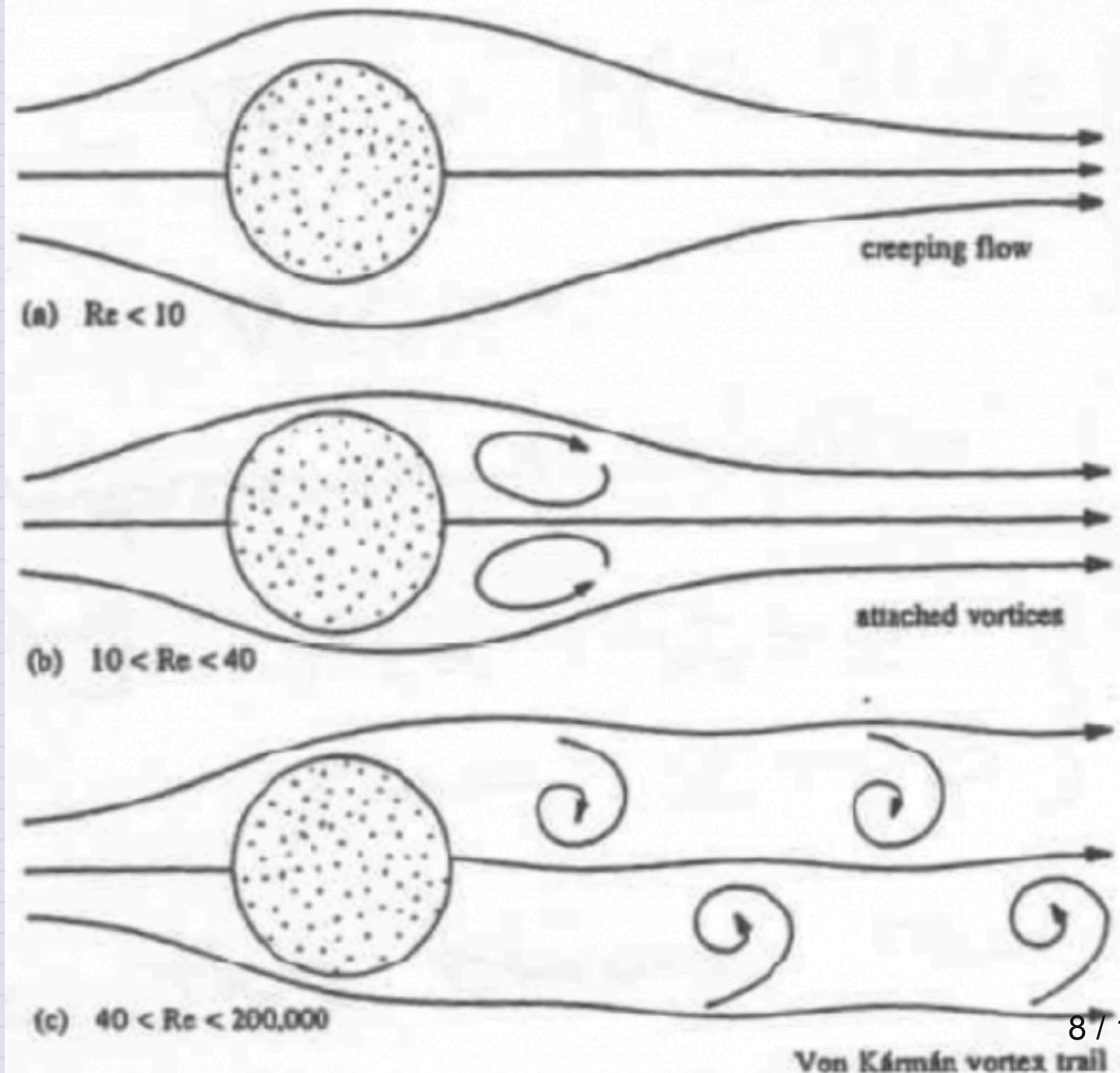
What is your experience with modeling?

Take 2-3 min to think about your prior physics classes

- What models have you used? What makes that a model?
- What made a that model good or not so good?
- What kinds of things could you do to make a better model?

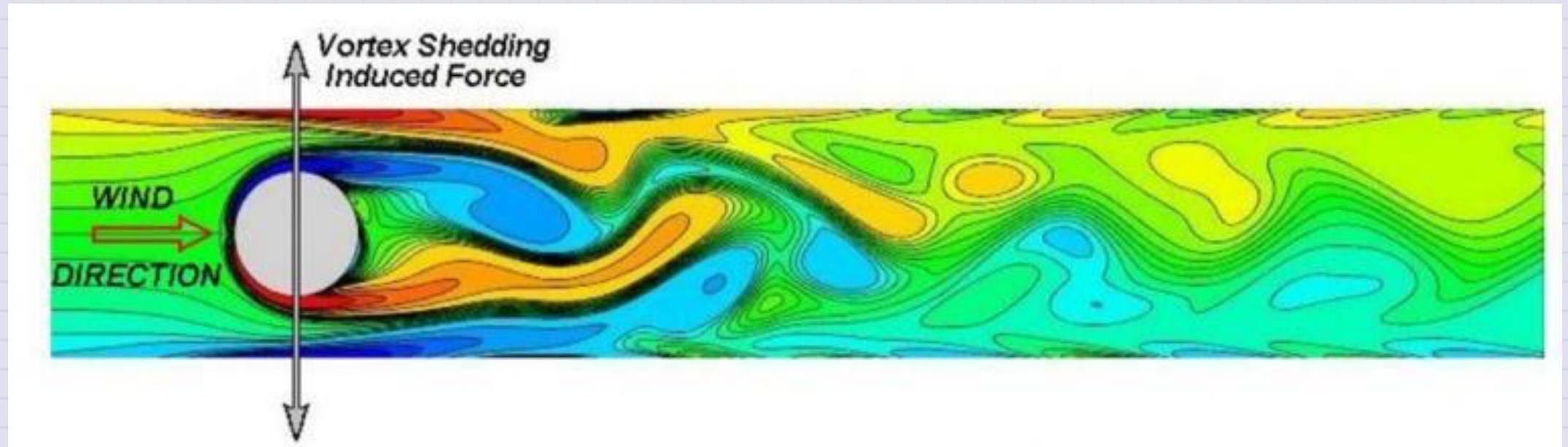
Vortex Shedding

- At higher Reynolds numbers, flow around objects becomes unstable.
- This instability can lead to the formation of vortices.
- This "shedding" of vortices can lead to vibrations and noise.



Model of vortex shedding behind a cylinder

- Controlling vortex shedding is important in many engineering applications.



Giosan, Ioan, and P. Eng. **"Vortex shedding induced loads on free standing structures"**
Structural Vortex Shedding Response Estimation Methodology and Finite Element Simulation 42
(2013).

Renewables: Wind Turbines

Thorntonbank Wind Farm

North Sea off the coast of Belgium

Notice the cylindrical shape of the support structure.



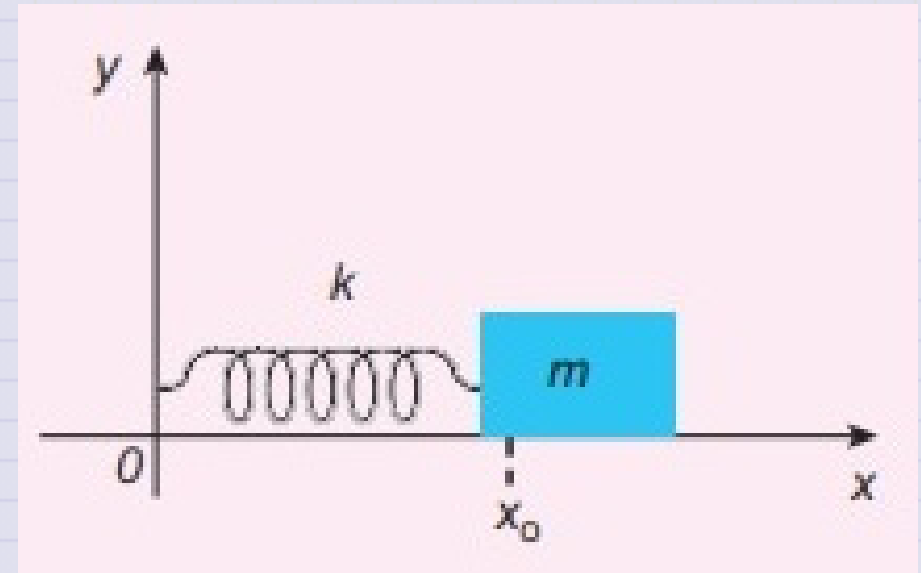
Clicker Question 6-1

The simple harmonic oscillator (SHO) is a useful model for many systems.

$$m\ddot{x} = -kx$$

Assuming the **restoring force is anti-symmetric* about the equilibrium position, what would be next term in the force model for a spring mass system?

1. $\sim x^2$
2. $\sim x^3$
3. $\sim x^4$
4. $\sim x^5$



Clicker Question 6-2

Assuming a **linear model** for Air Resistance $\sim bv$, we obtained this EOM for a falling ball:

$$\ddot{y} = -g + \frac{b}{m}\dot{y}$$

What happens when $\ddot{y} = 0$?

1. The ball stops moving ($v = 0$).
2. The ball reaches a velocity of mg/b .
3. The ball reaches a terminal velocity.
4. I'm not sure.

Clicker Question 6-3

For the system of **Linear Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

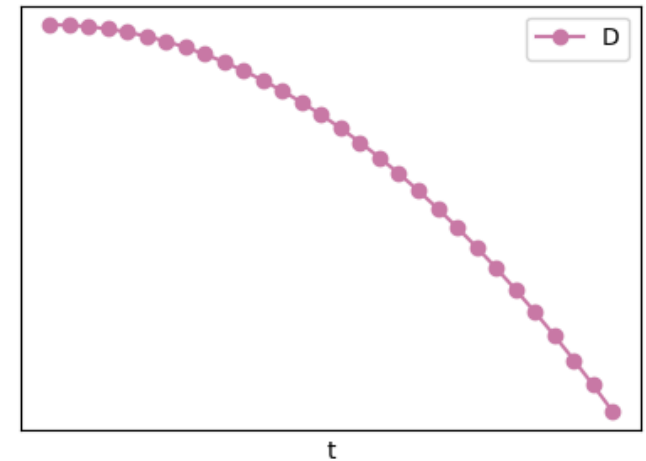
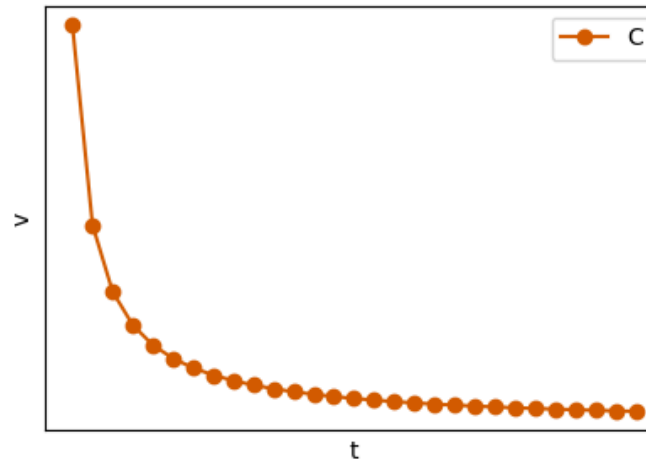
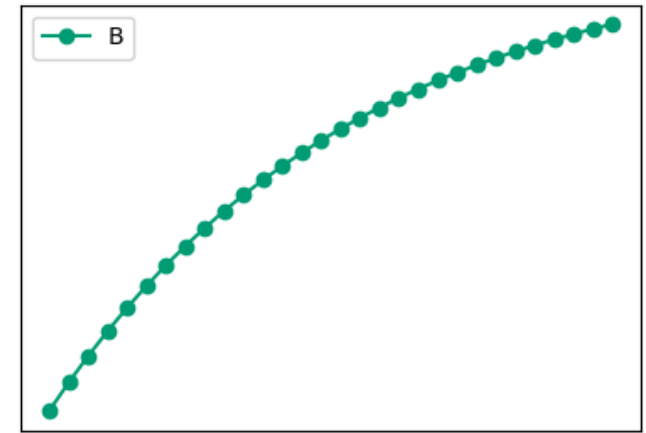
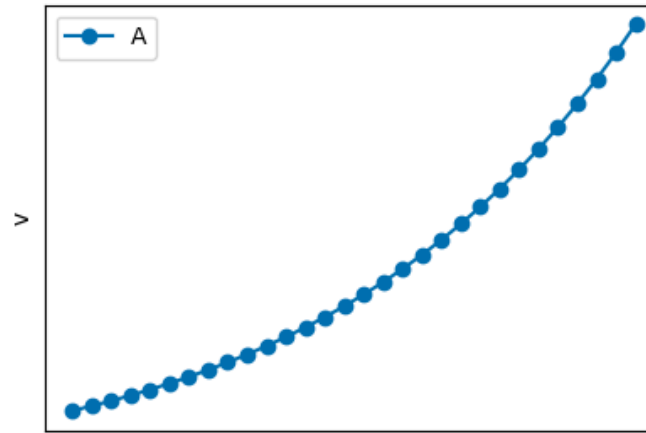
$$v(t) = v_{term} \left(1 - e^{-\frac{Ct}{m}} \right)$$

where $v_{term} = \sqrt{\frac{mg}{C}}$.

CQ 6-3

Which sketch could be correct for the velocity of the ball?

$$v(t) = v_{\text{term}}(1 - e^{-\frac{Ct}{m}})$$



Clicker Question 6-4

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = (mg/D)^{1/2}$. Do the units make sense? What are the units of $[gt/v_{term}]$?

1. Yes, the units for $[gt/v_{term}]$ are m/s ; both sides have the same units.
2. No, the units for $[gt/v_{term}]$ are m/s ; each side has different units.
3. Yes, the units for $[gt/v_{term}]$ are unit-less; both sides have the same units.
4. No, the units for $[gt/v_{term}]$ are unit-less; each side has the different units.

Clicker Question 6-5

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = \sqrt{mg/D}$. What happens when $t \rightarrow \infty$?

1. The object stops moving.
2. The object travels at a constant velocity.
3. The object travels at an increasing velocity.
4. The object travels at a decreasing velocity.
5. I'm not sure.